

Debugging Strategies

Project: Transparent Book Recommender

1. Debugging Strategies

Teach the learning cycle

Primary target: Design revealing tests for a function

Session · Sequence and minutes · Resources

Session: Session 1 · Understand · 45 min

Sequence and minutes: 0–7: recall and prerequisite; 7–15: inspect the concrete case and predict; 15–30: teacher models the worked trace; 30–40: learners explain a step using the text and visual; 40–45: check the core idea.

Resources: Sections 1–6; record the brief recall in the workbook.

Session: Session 2 · Practise · 45 min

Sequence and minutes: 0–5: retrieve the procedure; 5–15: W2 completion practice; 15–23: explain and check with a partner; 23–30: each learner answers the misconception check; 30–40: targeted reteaching or a harder variation; 40–45: prepare the investigation.

Resources: Sections 7–10; use the matching workbook W2 and misconception check.

Session: Session 3 · Investigate and transfer · 45 min

Sequence and minutes: 0–5: retrieve the decision rule; 5–20: investigate or trace; 20–30: start the independent case; 30–38: connect the project; 38–43: act on feedback; 43–45: plan the return check.

Resources: Sections 9 and 11–14. Give extra time to finish the full task set; use Section 15 when the week includes a coding lab.

This is a starting allocation of 135 minutes, not a validated duration. Preserve all named topics. Schedule extra time for connected examples, the full investigation, independent cases and project building. Repeated copies of a case share one recorded attempt; they are not new evidence.

Case exposure check: this source example appears in ai-2/6, python-bridge/12. Reuse a familiar case for

recall or modelling; verify that the case used to assess transfer has not already received feedback.

Retrieval answer guidance

Loops and Stopping

Without opening the old lesson, explain one key idea from Loops and Stopping. Give an example and a mistake that the idea helps you avoid.

Look for: Trace variable values before and after each loop iteration

Reference example, not the only acceptable wording: 2, 3 and 4; the stop value is excluded. The body runs three times.

Branching Logic

Without opening the old lesson, explain one key idea from Branching Logic. Give an example and a mistake that the idea helps you avoid.

Look for: Trace the selected branch in an if elif else chain

Reference example, not the only acceptable wording: if tests negative first; elif checks ≤ 12 only after that fails; else handles the rest. Unindented print runs after any selected branch, whose assignment gives label a value.

Use the student module and workbook section with this exact week title. The worked case, independent question and answers below form one aligned lesson sequence.

What the learner must demonstrate

Intended score is 3 per matching trait. Bug: score starts at 3 and adds 3 for each match. Give expected and actual results for 0,1,2 matches and repair the cause.

Concept to explain

Good debugging isolates a cause with controlled tests.

Term · Meaning

Term: bug

Meaning: A defect that makes a program behave differently from its intended specification.

Term: debug

Meaning: Find and repair the cause of a program's incorrect behaviour.

Term: controlled test

Meaning: A test with specified inputs, conditions and expected measurements.

Term: retest

Meaning: Run a test again after a change or under specified new conditions.

Prepare the class

Use the supplied scenario, student illustrations and blank paper. For numeric work, offer counters or a calculator after learners show the setup. Fictional records are supplied; personal data and paid accounts are unnecessary.

Allow 80 minutes: recall 8; explanation 12; worked demonstration 15; guided practice 20; independent assessment 15; project and reflection 10. Extend the lab into a second session when needed.

Preparation for the connected explanation

Teach Week 12 as a two-session investigation using fictional data. A function can pass one example by coincidence. Start with a written rule, calculate expected results independently, and include ordinary, zero and boundary cases. We will test a subtraction function under the contract $0 \leq \text{used} \leq \text{capacity}$. Assess the three stated outcomes through the independent task, not by copying the worked result.

Readiness answer

7.

Python 3 editor with a fresh local practice file

Paper trace table and the student workbook

Fictional input records from this lesson; no real student identifiers

End goal for the pictorial case

By the end, I can test a function with expected values that distinguish a particular bug.

This added worked case illustrates the week's main idea. Retain all original and connected topics; schedule its practice within the teaching cycle.

2. Explain the mechanism

Explain the mechanism

Input → Process → Output

Trace the information. At each stage, ask what changed and how you could test it.

Which stage could fail even when the other two are correct? Explain:

Predict first. Then check each step. A correct answer without visible reasoning is incomplete evidence.

1. Observed: wet-weather case recommends sandals.
2. Divide: input correct? yes. rain branch reached? no.
3. Cause: condition says rain = false instead of true.
4. Fix that condition only; rerun the original case and a dry-weather case.

Recalculate, retrace, or restate the reasoning in your own representation:

CHECK Did the evidence answer the exact question? Did the total, path, or source record stay consistent?

Concrete teaching case

Award two points per matching trait. A bug awards two for every trait.

Test one match and one nonmatch.

Exact explanation to model

Expected 2; buggy result 4. Put the addition inside the match condition. Test zero, one and two matches for 0,2,4.

Explain the added worked example

A function can pass one example by coincidence. Start with a written rule, calculate expected results independently, and include ordinary, zero and boundary cases. We will test a subtraction function under the contract $0 \leq \text{used} \leq \text{capacity}$.

Each parenthesized group is a tuple, an ordered group of values. The loop unpacks each group into three names. Equality compares the actual result to the independently calculated expected result. The worked script prints True three times. An assert actual == expected could instead stop at the first mismatch with AssertionError.

Key terms

Term · Meaning

Term: Test case

Meaning: Inputs and an expected result

Term: Boundary

Meaning: An edge of a permitted range

Term: Regression

Meaning: A previously working case that fails after a change

Term: Assertion

Meaning: A check that raises an error when its condition is false

Model the reasoning aloud

Use the complete worked example below before asking learners to complete W2. Reveal a small part, invite a prediction, then explain the deciding step. These teacher notes contain solutions; do not reveal the independent case answer during modelling.

Would replacing the body with return 5 pass every worked case? Identify the first revealing test.

Thinking move · Teacher prompt

Thinking move: Plan

Teacher prompt: What is given? What is the question? Which procedure fits these facts?

Thinking move: Do

Teacher prompt: Point to each input and intermediate step in the worked trace. Name the rule that allows the next step.

Thinking move: Check

Teacher prompt: Does the output answer the question? Which assumption or limitation prevents a larger claim?

Use the diagram and verbal explanation together. Ask learners to match a sentence to a specific visual step; do not assign learners fixed visual or auditory learning styles.

A pictorial worked case: Debugging Strategies

A fictional shelf function should count books in complete bundles plus loose books.

The facts we will use

Correct definition: shelf(bundle_count,loose) returns bundle_count*6+loose.

Faulty copied body returns (bundle_count+loose)*6. Predict tests before running.

AI ACADEMY / AI-2/6

Expected outputs versus a specific bug

	Bundles	Loose	Correct	Faulty
Test 1	2	1	13	18
Test 2	0	3	3	18
Test 3	2	0	12	12

Read a cell using both its row and column labels.

Debugging Strategies

Start with the supplied facts

Follow each arrow. Ask which fact allows the next step.



1. Expected normal test

For (2,1), correct result is $2 \times 6 + 1 = 13$.



2. Faulty trace

Copied code gives $(2+1) \times 6 = 18$.



3. Zero-bundle test

For (0,3), correct 3; copied code 18.



4. Nonrevealing test

For (2,0), both return 12; a pass does not locate every bug.

CHANGE ONE THING

Repair the body to `bundle_count*6+loose`.

Explain the changed result using the matching lesson.

Read the picture one step at a time

Step · What happens and why

Step: 1. Expected normal test

What happens and why: For (2,1), correct result is $2 \times 6 + 1 = 13$.

Step: 2. Faulty trace

What happens and why: Copied code gives $(2+1) \times 6 = 18$.

Step: 3. Zero-bundle test

What happens and why: For (0,3), correct 3; copied code 18.

Step: 4. Nonrevealing test

What happens and why: For (2,0), both return 12; a pass does not locate every bug.

Match each pictured step to the supplied facts. This is a worked teaching case, not an independent assessment.

Why this result follows

Test cases need not all expose the same error. Here loose books must be added after multiplying bundles. A zero-loose case cannot distinguish the two formulas because multiplying zero loose books changes nothing. Keep expected and actual columns so an observed output does not quietly become its own expected answer.

What this example does not establish

These three tests check a specified formula and bug; they do not prove correct handling of every type or invalid request.

Change one thing in the pictured case

Change: Repair the body to $\text{bundle_count} * 6 + \text{loose}$.

Prediction: Expected results for (2,1),(0,3),(2,0) are 13,3,12.

Reason: Only bundles are groups of six; loose books are counted individually.

Ask while showing the picture

Cover the result. Ask learners to name the inputs and predict the next step. Uncover one step, connect the icon to the actual procedure, and explain its evidence. Then change the supplied condition and compare. The picture is a representation; it does not document execution of a real AI model.

3. Demonstrate and guide practice

Demonstrate and guide practice

Board sequence

Write the given facts: Award two points per matching trait. A bug awards two for every trait.

Write the question to explain: Test one match and one nonmatch.

Reveal after predictions: Expected 2; buggy result 4. Put the addition inside the match condition. Test zero,

one and two matches for 0,2,4.

Keep for an independent attempt: A loop should run four times but runs five. You change its colour and stopping condition at once. Which single change is relevant to test first?

Classroom dialogue

Ask: Which facts are given, and which would we have to assume? Learners mark relevant details before discussing answers.

Say: Predict first. Show the rule, information path or calculation that would produce the result. A correct guess is not yet an explanation.

Reveal reasoning one step at a time. Explain why a tempting alternative fails. Write units and denominators for arithmetic, trace conditions for decisions, and identify supporting sources for claims.

In pairs, one learner solves while the other checks evidence; exchange roles. Remove the worked answer before independent assessment.

Model these guided questions

It passes (8, 3) but fails (8, 8), returning 5 instead of 0. Exit: No. Negative counts are outside our nonnegative-input contract and were not tested; rejection behavior must first be defined.

Case-specific teaching sequence

Stage · Teacher action

Stage: Session 1 start to 7 minutes

Teacher action: Retrieve the prerequisite and explain the purpose in the opening section.

Stage: 7 to 20 minutes

Teacher action: Read the worked example and diagram; ask learners to trace each step aloud.

Stage: 20 to 33 minutes

Teacher action: Complete W1 and W2, then compare predictions with observed results.

Stage: 33 to 45 minutes

Teacher action: Collect the guided prediction before running; discuss a mismatch and its cause.

Stage: Session 2 start to 8 minutes

Teacher action: Retrieve the main rule without showing the previous answer.

Stage: 8 to 20 minutes

Teacher action: Use W3 to diagnose and repair the stated failure; rerun an earlier case.

Stage: 20 to 35 minutes

Teacher action: Collect W4 independently, including code or plan, predictions and explanations, before hints.

Stage: 35 to 45 minutes

Teacher action: Discuss W5 and the exit; use the fresh retry when support was needed.

Reduce help deliberately

Completion task: A faulty function always returns 4. Which of your three cases pass and fail?

Teacher check: Only (6, 2) passes. The full and unused cases fail.

Start by identifying the given inputs together. Leave the intermediate steps blank. If needed, model one step, then withdraw that help on the next step. Move to a full trace only when the partial trace is understood. For partners, alternate explainer and checker; collect a response from every learner before discussion.

Individual reasoning check

Use the worked case for Testing and debugging functions. Proposed explanation: "A successful run proves the conclusion". Decide whether this explanation is supported. Point to the step or supplied fact that decides; explain your repair if needed.

Expected correction: Passing this small suite supports these particular cases. It neither proves every allowed input correct nor checks invalid input handling. A test outside the contract needs a separately specified expected behavior.

Collect explanations, not only yes/no votes. If a misunderstanding is common, re-model the relevant step for the class. If it affects a few learners, give a short targeted group explanation while others solve a more demanding variation.

Pictorial practice question

Which test most directly isolates the handling of loose books, and what should it produce?

Reasoned teacher answer

(0,3) isolates loose books: expected 3. The faulty body gives 18.

Changed-case reasoning: Expected results for (2,1),(0,3),(2,0) are 13,3,12. Only bundles are groups of six; loose books are counted individually.

Collect each learner's explanation before discussion. Use the first missing connection between input, deciding step and result to choose support. This question is worked-case practice, not fresh mastery evidence.

4. Run the weekly evidence lab

Run the weekly evidence lab

Words You Can Use to Reason

A definition matters when you can recognize the idea, use it, and contrast it with a near-miss.

Predict → Test → Record → Explain

Keep expected and actual results separate. Surprises are useful data when recorded honestly.

Use normal, boundary, and unexpected test cards. Mark the first step where expected and actual behavior differ; change one cause; rerun all three.

What did the hardest test teach you that the normal test did not?

Make the lab runnable

Use the concrete case as the shared normal case and the independent question as a second case. Each pair creates a boundary or missing-information case and predicts its result before testing. Keep the expected result separate from an observed mistake.

Record · What to capture

Record: Input or evidence

What to capture: Exact data, instruction or claim

Record: Expected result

What to capture: Answer or safe fallback declared before the run

Record: Observed result

What to capture: Actual result, including errors

Record: Explanation

What to capture: Rule, calculation or evidence explaining the outcome

Record: Next test

What to capture: One justified change and a new prediction

Diagnose and reteach

For guessing, return to given facts and request a trace. For vocabulary without application, use one example and one non-example. For arithmetic errors, check the setup before the calculation. Finish with a different uncoached case before recording mastery.

Address these additional misconceptions

A successful run proves the conclusion: Passing this small suite supports these particular cases. It neither proves every allowed input correct nor checks invalid input handling. A test outside the contract needs a separately specified expected behavior.

Matching the example is enough: Require the new W4 case, explicit predictions and an explanation. Fresh retry: bundles * 3 + singles with (3, 1), (0, 1), (3, 0) expects 10, 1, 9. A constant return 10 passes only the first.

A convenient shortcut preserves meaning: A faulty return capacity + used gives 11, 16 and 8. The unused case passes even though the rule is wrong. Repair the operator to subtraction and rerun all three cases. Never change an expected result merely to match faulty output.

Match the support to the first error

What the response shows · Next teaching action

What the response shows: Missing or misread facts

Next teaching action: Read the case aloud, mark supplied versus unknown facts, and let the learner restate it.

What the response shows: Wrong procedure

Next teaching action: Return to the deciding step in the worked case. Contrast it with the proposed incorrect explanation.

What the response shows: Arithmetic or implementation error

Next teaching action: Trace intermediate values or lines; preserve the first mismatch and retry that step.

What the response shows: Correct result without a reason

Next teaching action: Ask for a trace and a changed case before marking independent understanding.

What the response shows: An unsupported wider claim

Next teaching action: Ask which supplied evidence supports the claim and which further observation would be needed.

Offer an oral explanation, drawing, enlarged visual or fewer examples when that removes a reading/writing barrier. Keep the same reasoning goal. Stretch secure learners by changing a boundary or assumption and asking them to defend the effect, not simply by assigning more of the same questions.

5. Assess and explain the answers

Assess and explain the answers

Worked case answer

Expected 2; buggy result 4. Put the addition inside the match condition. Test zero, one and two matches for 0,2,4.

Independent question

A loop should run four times but runs five. You change its colour and stopping condition at once. Which single change is relevant to test first?

Expected independent answer

Test the stopping condition first while preserving the other code. Trace the counter at the boundary; changing colour provides no evidence about repetition.

What counts as evidence

Criterion · 0 points · 1 point · 2 points

Criterion: Result

0 points: Incorrect or absent

1 point: Partly correct

2 points: Correct with necessary conditions

Criterion: Reasoning

0 points: No supporting trace

1 point: Some steps

2 points: Clear mechanism, calculation or source support

Criterion: Limit

0 points: Overclaims certainty

1 point: Vague limitation

2 points: Relevant uncertainty or missing fact

Criterion: Transfer

0 points: Copies without adapting

1 point: Adapts with prompts

2 points: Solves a changed case independently

Use 6/8 as a classroom checkpoint, with 2 required for both Result and Reasoning. This is a proposed teaching rule, not a validated certification threshold. Do not count a coached answer as independent mastery; correct unsafe or unsupported claims regardless of total.

Feedback

Ask the learner to find the evidence that challenges this claim: “Any visible change helps diagnose the cause.” Keep the first attempt and an uncoached retry separate.

Original transfer answer

Expected 0,3,6; actual 3,6,9. Initialise score to 0, keeping the per-match addition. The zero-match case isolates the initialisation error. Rerun all three tests; changing only a displayed result would leave the cause intact.

Explanations for the additional workbook examples

W1 Write expectations

Give expected results for remaining(6, 2), remaining(6, 6) and remaining(6, 0).

4, 0 and 6.

Apply the specified subtraction rule to each pair: $6 - 2 = 4$, $6 - 6 = 0$, and $6 - 0 = 6$. Label them ordinary, full and unused cases. These expected answers come from the contract before the program is run.

W2 Find a revealing case

A faulty function always returns 4. Which of your three cases pass and fail?

Only (6, 2) passes. The full and unused cases fail.

Compare the faulty actual result 4 with each independently expected result. It matches the ordinary case expecting 4, but not the full case expecting 0 or unused case expecting 6. One successful case therefore does not establish that the subtraction rule was implemented.

W3 Repair and retest

A faulty function adds capacity and used. For (6, 2), state actual and expected. Repair it and name a regression case.

Actual 8, expected 4. Subtract used; rerun a prior case such as (6, 0), expecting 6.

W4 Test a new function independently

Define `stock(packs, loose)` to return `packs * 4 + loose`. Create expected tests for (2, 3), (0, 3), (2, 0), then show which detects a faulty `packs + loose` body.

Expected 11, 3, 8. Faulty results 5, 3, 2; first and third reveal it. Use independent expectations in the test loop.

W5 Limit the conclusion

All three stock tests pass. Does that prove correct handling of negative counts?

No. Negative counts are outside our nonnegative-input contract and were not tested; rejection behavior must first be defined.

Assess independent transfer

Selected case: Define `stock(packs, loose)` to return `packs * 4 + loose`. Create expected tests for (2, 3), (0, 3), (2, 0), then show which detects a faulty `packs + loose` body.

Selected case answer

Expected 11, 3, 8. Faulty results 5, 3, 2; first and third reveal it. Use independent expectations in the test loop.

Dimension · Evidence required

Dimension: Explanation

Evidence required: Names the applicable idea or procedure and ties it to the supplied facts.

Dimension: Trace and result

Evidence required: Preserves the given inputs and shows the relevant intermediate steps; distinguishes a prediction from an observed execution.

Dimension: Limits

Evidence required: States the assumptions, missing evidence or conditions under which the result would change.

Record each dimension as not yet, with support, or independent. Accept a valid alternative procedure with a defensible trace. Do not penalise an honestly recorded model failure if the learner correctly explains its evidence and limits.

Feedback that the learner uses

Name the earliest unsupported step and give one action to repair it. Have the learner make that repair now and explain why. Then assess the relevant skill on a fresh, fully specified case. Confirm it was not previously practised with feedback; solve and check the new case before assigning it. A follow-up design prompt is a starting brief, not automatically an equivalent test.

6. Connect and extend learning

Connect and extend learning

Transparent Book Recommender

When a ranking surprises you, inspect feature values, weights, arithmetic, and sorting separately.

Keep the decision, input, result, explanation and remaining limitation in the project evidence record. Retrieve this record next week rather than restarting the project.

Support and stretch

Trace one record or condition at a time with counters and labelled columns. Stretch: introduce a tie, boundary or missing field and declare its handling policy before rerunning.

Exit reflection

Ask for one decision the learner can now justify and one situation requiring more evidence or human review. Confidence ratings are separate from correct independent performance.

Bridge to the next week

Next: Week 7: Rows, Columns, and Data Types. Retrieve today's evidence and identify what carries forward and what is new.

Support and fresh reassessment

Provide a blank inputs expected actual table. Extension: use assert for the same cases and explain why only the first failure is reported.

Independent W4 evidence: Expected 11, 3, 8. Faulty results 5, 3, 2; first and third reveal it. Use independent expectations in the test loop. Record whether the learner needed no help, a hint or explanation. Fresh retry: bundles * 3 + singles with (3, 1), (0, 1), (3, 0) expects 10, 1, 9. A constant return 10 passes only the first.

Next week: Rows columns and schemas. Reuse this week's explicit contracts and evidence.

Plan the delayed return

Suggested return points in this level: ai-2/7, ai-2/9, ai-2/12. Use actual classroom dates, adjusting for holidays and gaps. One, three and six teaching weeks are scheduling choices, not a claimed optimal spacing.

At a delayed check, ask for the idea from memory and a changed application. Compare with the earlier independent response and record support used. For a new level, revisit the previous capstone before checking the new prerequisites.

Record · Teacher evidence

Record: Learner code and date

Teacher evidence: _____

Record: Concept and case used; prior exposure

Teacher evidence: _____

Record: Explanation / trace / limits

Teacher evidence: _____

Record: Support and first repair

Teacher evidence: _____

Record: Changed case and delayed outcome

Teacher evidence: _____

Record: Next teaching action

Teacher evidence: _____

Use the evidence to choose reteaching or the next step. A completed page, confidence rating or attractive project does not by itself demonstrate understanding.

Project evidence from this case

Test the recommender's helper functions with normal, zero and boundary inputs before debugging the whole application.

7. Prepare, teach and assess

Teaching target: Design revealing tests for a function

Prepare the labelled worked-case picture, the matching workbook, fictional input cards and a place for individual responses. No paid AI account is required for the paper reasoning activities.

Session · Plan · Resources

Session: Session 1 · Understand · 45 min

Plan: 0–7: recall and prerequisite; 7–15: inspect the concrete case and predict; 15–30: teacher models the worked trace; 30–40: learners explain a step using the text and visual; 40–45: check the core idea.

Resources: Sections 1–6; record the brief recall in the workbook.

Session: Session 2 · Practise · 45 min

Plan: 0–5: retrieve the procedure; 5–15: W2 completion practice; 15–23: explain and check with a partner; 23–30: each learner answers the misconception check; 30–40: targeted reteaching or a harder variation; 40–45: prepare the investigation.

Resources: Sections 7–10; use the matching workbook W2 and misconception check.

Session: Session 3 · Investigate and transfer · 45 min

Plan: 0–5: retrieve the decision rule; 5–20: investigate or trace; 20–30: start the independent case; 30–38: connect the project; 38–43: act on feedback; 43–45: plan the return check.

Resources: Sections 9 and 11–14. Give extra time to finish the full task set; use Section 15 when the week includes a coding lab.

Before class, solve the supported practice: A faulty function always returns 4. Which of your three cases pass and fail?

Reasoned practice answer: Only (6, 2) passes. The full and unused cases fail.

Check every learner before discussion: Use the worked case for Testing and debugging functions.

Proposed explanation: “A successful run proves the conclusion”. Decide whether this explanation is supported. Point to the step or supplied fact that decides; explain your repair if needed.

Misconception repair: Passing this small suite supports these particular cases. It neither proves every allowed input correct nor checks invalid input handling. A test outside the contract needs a separately specified expected behavior.

Support: read the exact case aloud, enlarge the labelled picture, supply input cards and model only the first deciding step. Accept a spoken explanation or drawing; retain the reasoning goal.

Extension: Repair the body to `bundle_count*6+loose`. Require a prediction and a reason before comparing. Expected result: Expected results for (2,1),(0,3),(2,0) are 13,3,12. Reason: Only bundles are groups of six; loose books are counted individually.

Independent assessment: withhold the reserved answer until the learner has attempted the new case. Record support separately from correctness.

Criterion · Not yet · With support · Independent

Criterion: Explanation

Not yet: Names an answer without a deciding fact

With support: Uses a hint to link fact and decision

Independent: Explains the deciding fact in own words

Criterion: Trace or test

Not yet: Cannot connect input and result

With support: Completes steps with prompts

Independent: Traces or tests a changed case accurately

Criterion: Limits and revision

Not yet: Claims more than evidence supports

With support: Names a limit after a prompt

Independent: States a defensible limit and useful next test

Project checkpoint: Frame the need. Name the user, the problem, allowed inputs and a safe simple starting method.

Use the weekly teacher answer for topic-specific reasoning. A saved progress tick is an activity record; judge mastery from the learner's explanation and evidence.

8. Use feedback, labs and repair paths

Use the two formative questions as supported practice, not as independent mastery scores. Discuss the deciding evidence rather than accepting a guessed selection.

The feedback comes from the reviewed public worked case. Do not reveal the separately reserved independent teacher answer until the learner has attempted that case.

Ask students to predict before running code. Preserve errors and actual outputs; collect a changed test and an explanation. The branch-boundary and indexing lessons deliberately include failure cases.

Use the earlier worked-explanation links to address a specific missing building block, then return to the current question. Adapt this suggestion after observing the learner; it is not an automatic diagnosis.

Collect the completed-work portfolio as evidence. Record support separately from correctness, and assess explanation, trace/test and limitations. Progress ticks and quiz selections alone do not establish mastery.